

10 C At Singapore, $R = R_{\text{Earth}}$

$$\Rightarrow a_{\text{Singapore}} = R_{\text{Earth}} \omega^2$$

At Seoul, $R = R_{\text{Earth}} \cos 37.5^\circ$

$$\Rightarrow a_{\text{Seoul}} = R_{\text{Earth}} \cos 37.5^\circ \omega^2$$

$$\frac{a_{\text{Singapore}}}{a_{\text{Seoul}}} = \frac{R_{\text{Earth}} \omega^2}{R_{\text{Earth}} \cos 37.5^\circ \omega^2} = \frac{1}{\cos 37.5^\circ} = 1.26$$

$$dV = 322.5 \text{ (369.2)} = 46.7$$